Rapid Revision Summary

Last-minute review · condensed from 17 papers and 16 mark schemes, Jan 2019 – Jun 2026

THE PAPER

2 hours · 100 marks · 21–26 questions · calculator · ≈ 1.2 min per mark. Same content as Paper 1H, but more recurring decimals, proofs, constructions, graph work, 3-set Venn diagrams and distribution comparisons.

ON THE FORMULAE SHEET

S_n · quadratic formula · trapezium · sine/cosine rules · ½ab sin C · cone, sphere, cylinder, prism

Learn: arcs, sectors, circle theorems, index laws, bounds, intersecting chords.

TOP 12 MARK-LOSERS

- "Show/use algebra" with answer only = **0**.
- Recurring decimal by rule, not two equations.
- Proof with numbers instead of n.
- Construction arcs rubbed out.
- Graph solution without the drawn line.
- Conditional probability over the whole total.
- Comparison with no context or no values.
- "(1 − 15%)" instead of **0.85**.
- Proportion formula without **k**.
- Wrong region in a quadratic inequality.
- Circle-theorem angle without a reason.
- Simplifying a correct answer wrongly.

NUMBER

RECURRING DECIMALS

x = 0.5393…: 1000x = 539.39…, 10x = 5.39… ⇒ 990x = 534 ⇒ 89/165.

0.3787… = 25/66 · 0.6818… = 15/22.

- Indices: write numbers as prime powers; $(9x^4/16y^{10})^{-½} = 4y^5/3x^2$.
- Surds: $\sqrt{8} = 2\sqrt{2}$; $(5 - \sqrt{18})/(1 - \sqrt{2}) \times (1 + \sqrt{2})/(1 + \sqrt{2})$.
- LCM highest powers, HCF lowest.
- Bounds: density $3M/4r^3$ max uses $M_{max} = 265$, $r_{min} = 2.75$ ⇒ 9.56.

PERCENTAGES AND RATES

- Reverse two-stage: 687 700 ÷ (1.15 × 0.92) = 650 000.
- Overall: 1.15 × 0.92 = 1.058 ⇒ +5.8%.
- Compare: 45/500 = 9% vs 47/545 = 8.6%.
- Sale price 952 at 15% off ⇒ normal 1120, discount 168.
- km/h → m/s: × 5/18. 50 m/s = 180 km/h.
- Value for money: same currency, same unit.

ALGEBRA AND PROOF

PROOF

$(n + 1)^2 - n^2 = 2n + 1$ ⇒ odd.

$(2n + 1)^2 + (2n + 3)^2 = 8(n^2 + 2n + 1) + 2$ ⇒ remainder 2.

- Subject twice: $p = (ac + 8)/(3 + c)$ ⇒ $c = (8 - 3p)/(p - a)$.
- $3x^2 - 6x + 5 = 3(x - 1)^2 + 2$.
- $28 + 24x - 6x^2 = 52 - 6(x - 2)^2$ → max (2, 52).
- Formula form $(-4 \pm 2\sqrt{39})/10$ ⇒ match $-b/2a$ and $b^2 - 4ac$.

EQUATIONS AND INEQUALITIES

CIRCLE + LINE

$y = 7 - 2x$, $x^2 + y^2 = 34$ ⇒ $5x^2 - 28x + 15 = 0$ ⇒ (5, −3), (0.6, 5.8).

- $3y^2 + 4y - 32 > 0$ ⇒ $(3y - 8)(y + 4) > 0$ ⇒ $y < -4$ or $y > 8/3$.
- Area inequalities: lengths > 0.
- Regions: draw lines, label R.

SEQUENCES AND FUNCTIONS

- a = 10, $u_{20} = 86$ ⇒ d = 4; $S_N = 5194$ ⇒ N = 49.
- gf(x) = g of f. $f^{-1}(7)$: solve f(x) = 7.
- $g = 14 - (x - 3)^2$, $x \geq 3$ ⇒ $g^{-1}(x) = 3 + \sqrt{(14 - x)}$, $x \leq 14$.

GRAPHS

- Match: x^2 U · $-x^2$ ∩ · −5/x 2nd/4th quadrants · cubic · sin · exponential.
- Solve by line: rearrange so plotted curve = line. $x^2 - x - 1 = 0$ with $y = 2x^2 - 4x - 1$ ⇒ draw $y = -2x + 1$.
- Gradient: tangent ≥ 3 cm, rise ÷ run.
- $y = a \sin x + b$: amplitude a, centre y = b.
- Perpendicular: $m_1 m_2 = -1$; $5y + 2x = 7$ ⇒ ⊥ gradient 5/2.

CALCULUS

- Gradient 11 on $y = x^3 - 16x + 7$ ⇒ $x = \pm 3$ ⇒ b = −47, 61.
- Direction reverses when v = 0: $x = 4t^3 - 27t + 8$ ⇒ t = 1.5, a = 36.
- Turning points: dy/dx = 0.

GEOMETRY

CIRCLES

Centre = 2 × circumference · semicircle 90° · same segment · cyclic 180° · tangent ⊥ radius · equal tangents · alternate segment.

Intersecting chords: PT × TQ = ST × TR ⇒ 3PT = 48 ⇒ r = 9.5.

- Obtuse sine-rule angle: 180° − calculator.

- Quadrilateral area: two triangles.
- 3D: right-angled triangle with the plane.
- Back bearing ± 180°.
- Perpendicular bisector and angle bisector: keep arcs.

MENSURATION AND SIMILARITY

- $4\pi r^2 = 49\pi \Rightarrow r = 3.5 \Rightarrow V = 180$.
- Area 25 : 64 ⇒ length 5 : 8 ⇒ volume 125 : 512; difference 541.8 ⇒ $V_B = 175$.
- Area 200 : 450 ⇒ 2 : 3 ⇒ volume × 27/8: 672 → 2268.
- Frustum = big cone − small cone.

VECTORS

- Two routes, equate a and b coefficients.
- Trapezium: CD = (1/3)AB ⇒ parallel.
- $|(x, y)| = \sqrt{(x^2 + y^2)}$.

STATISTICS

- Sixth value: 6 × 401 − 5 × 398 = 416.
- CF: median n/2; quartiles n/4, 3n/4; “more than” = total − reading.
- Histogram: f = density × width; conditional uses only the given part.
- Compare: median (on average…) + IQR (more consistent…), with values.

PROBABILITY AND SETS

- At least one red left (3R, 9G, take 3): 1 − (3/12)(2/11)(1/10) = 219/220.
- P(≥1 red) = 39/64 ⇒ P(G) = 5/8 ⇒ exactly one red = 15/32.
- 3-set Venn: centre → pairs − centre → singles → none.
- $P(G \mid C) = n(G \cap C) \div n(C)$.
- $A \cap C'$ in A not C; $(A \cup B)'$ outside both.